MACHINE LEARNING CASE STUDY

Classifying the Cosmos

Predicting whether a sky object is a galaxy, star, or quasar from telescope light measurements

SDSS photometric dataset • End-to-end analytics workflow

THE BUSINESS PROBLEM

Why does this problem exist?

Modern sky surveys photograph millions of objects every night. There are not enough astronomers alive to classify them by hand. The question is operational, not just scientific:

Can we automatically sort each observed object into **galaxy, star, or quasar** using only the measurements a telescope already captures?

- **Scale** — Automate what humans cannot do by hand across millions of objects
- **Triage** — Flag the interesting objects worth expensive follow-up observation
- **Consistency** — Apply the same criteria every time, without fatigue or bias

THREE POSSIBLE ANSWERS

Galaxy

Billions of stars bound together

Star

A single nearby sun-like body

Quasar

A black hole devouring matter

HOW WE'LL GET THERE

The analytics workflow

01
Exploratory
Understand the raw data

02
Descriptive
Summarise what's there

03
Feature Engineering
Build smarter inputs

04
Predictive
Train & compare models

05
Evaluate
Test honestly for traps

Each stage answers a different question — and each one guards against a different way of fooling yourself.

KNOW YOUR INPUTS

What the telescope gives us

Each row is one observed object. Its measurements fall into four groups:

Five brightness bands

`u g r i z`

The same object photographed through five colour filters, ultraviolet to infrared. The core signal.

Redshift

`how far it is`

How stretched the object's light is by cosmic expansion. Extremely predictive — and a trap we'll return to.

Sky position

`alpha delta`

Where the object sits on the celestial sphere. Should not matter for classification.

Identifiers

`IDs & run info`

Bookkeeping about which telescope run captured it. No predictive value — dropped.

STAGE 01 — EXPLORATORY ANALYTICS

Turning the lights on

Before modelling anything, interrogate the data. Exploratory analysis asks four blunt questions:

1 **Is the data even okay?**

Shape, missing values, duplicates, silly outliers like -9999 error codes

2 **What am I predicting?**

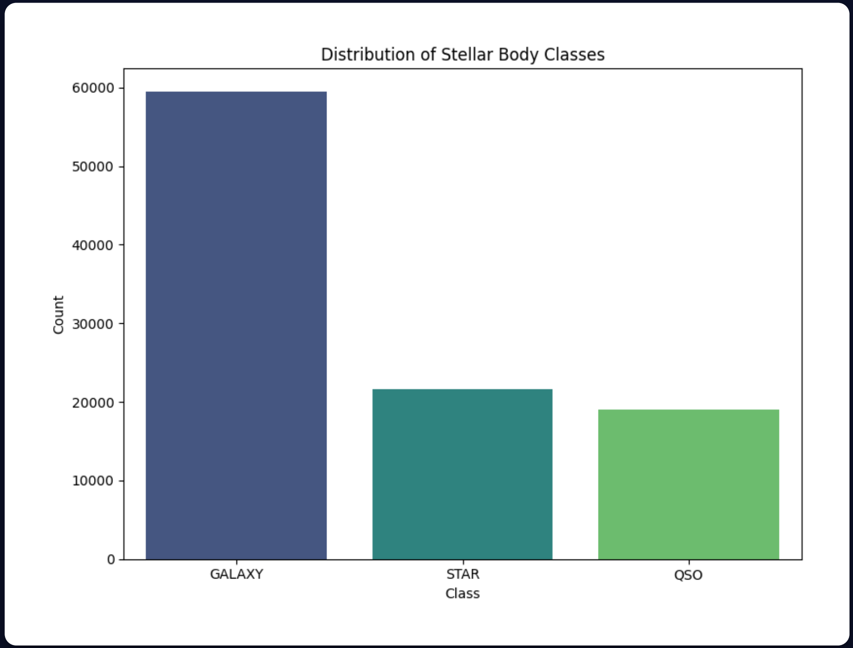
Check class balance — galaxies dominate (~60%), so that's the baseline to beat

3 **Are any features cheating?**

Redshift almost gives the answer away — flagged early

4 **Are inputs redundant?**

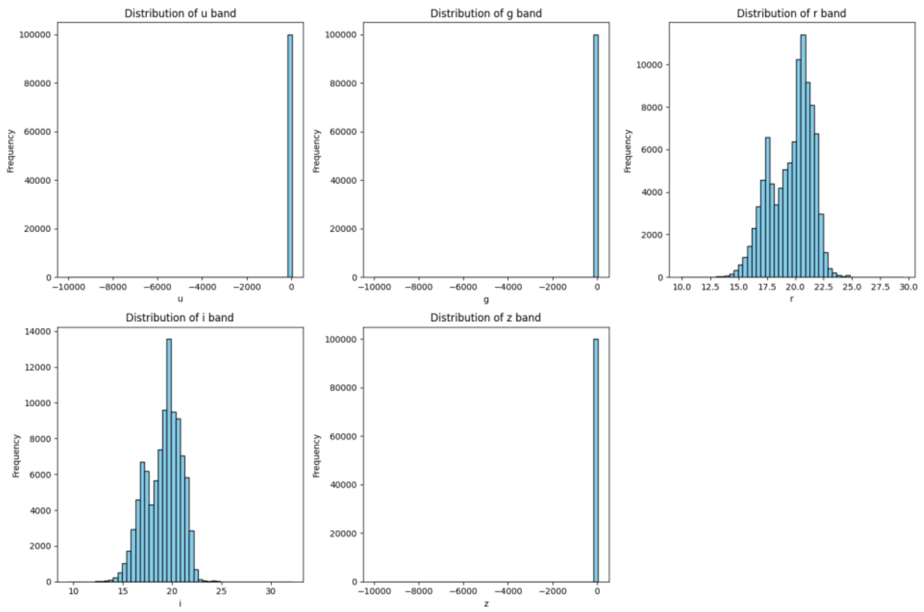
The five bands are all highly correlated with one another



Galaxies dominate at ~60% — the baseline any model must beat.

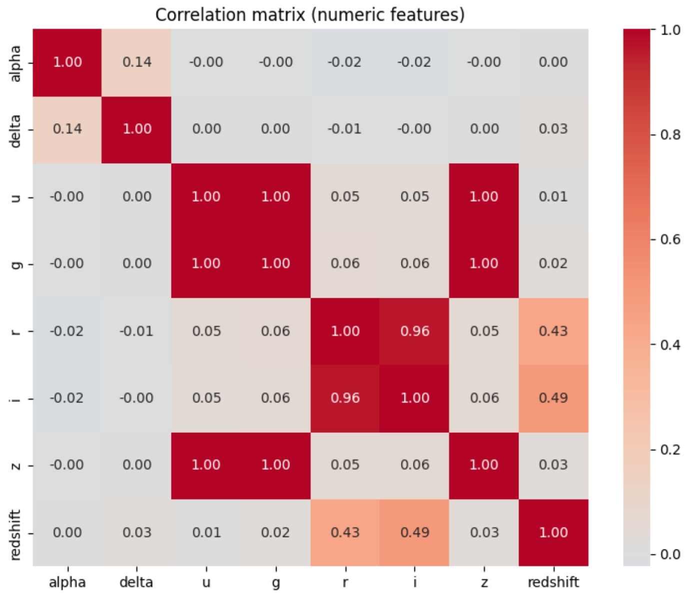
What the numbers actually say

Descriptive analytics summarises the data with charts — no prediction yet, just an honest portrait. Two views mattered most:



Raw band distributions

Note the huge spike near −9999 in u, g, z — SDSS's "bad measurement" sentinel, caught and handled.



Correlation matrix

u, g, z are near-perfectly correlated (1.00) — the redundancy that motivated feature engineering.

Building smarter inputs: colour

Raw brightness confuses distance with type. The fix — used by astronomers for a century — is to take differences between bands, called colour indices.

`u_g = u - g`

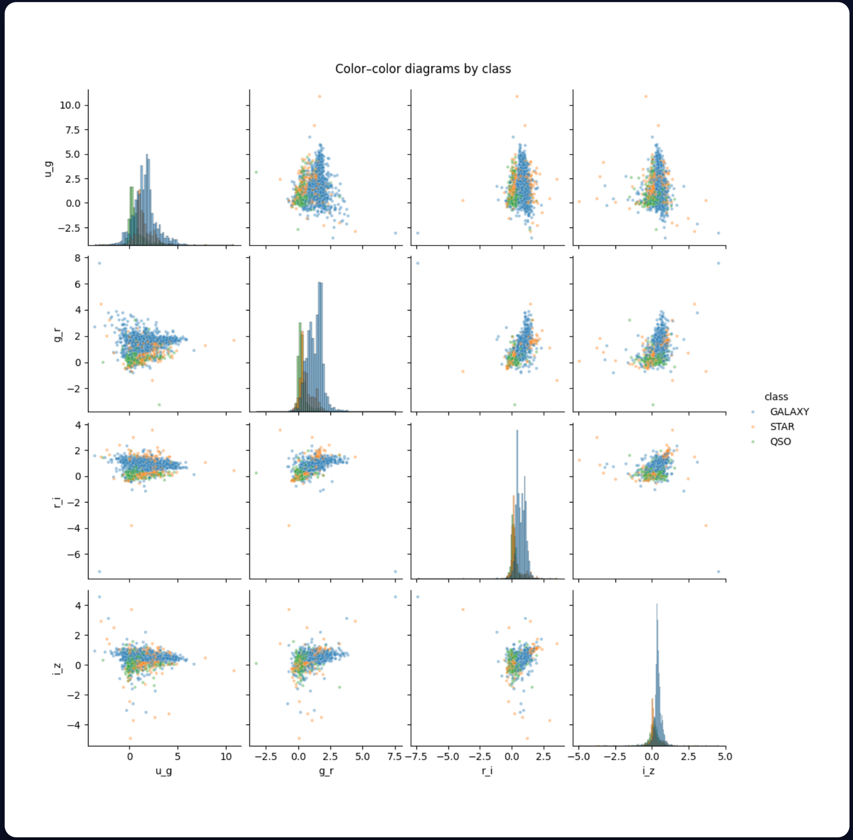
`g_r = g - r`

`r_i = r - i`

`i_z = i - z`

Why it works: a colour index captures the ***shape*** of an object's light, independent of how far or bright it is. Two stars of the same type share a colour even at different distances.

Sourced, not invented. You don't need to be a physicist — you notice the redundancy in the data, then look up how the field handles it, then confirm it empirically.



Colour–colour diagrams: the classes begin to separate in engineered feature space.

STAGE 04 — PREDICTIVE ANALYTICS

Two models, on purpose

We trained a simple linear model and a flexible ensemble — not to crown a winner, but to diagnose the shape of the problem.

Logistic Regression

The honest baseline

- Draws straight-line boundaries between classes
- Simple, fast, and fully interpretable
- You can read exactly what it learned from its coefficients
- Fails visibly when relationships are curved — which is useful

Random Forest

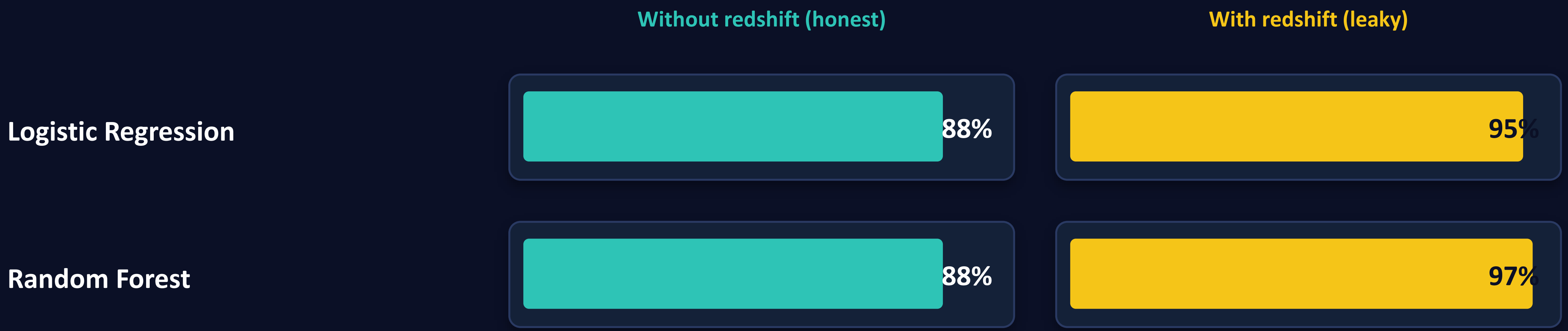
The flexible workhorse

- A committee of hundreds of decision trees that vote
- Captures curves and feature interactions naturally
- Ignores feature scale — no scaling headaches
- Reports which features it relied on most

STAGE 05 — EVALUATE HONESTLY

The result — and the trap

Each model was trained twice: without redshift, then with it. Redshift only comes from expensive spectroscopy, so a real image-only classifier can't use it. Including it is proxy data leakage.

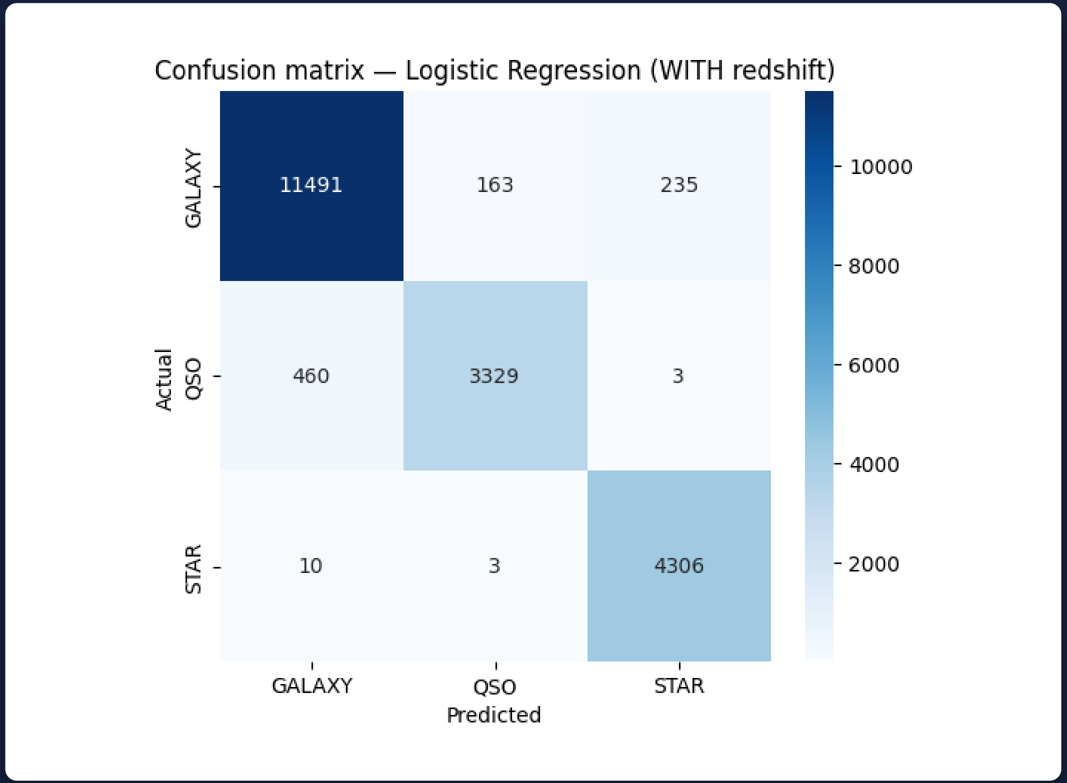


The 7–9 point jump is the leakage, quantified. For a real image-only pipeline the honest ceiling is ~88% — and a simple linear model reaches it. Feature quality mattered more than model choice.

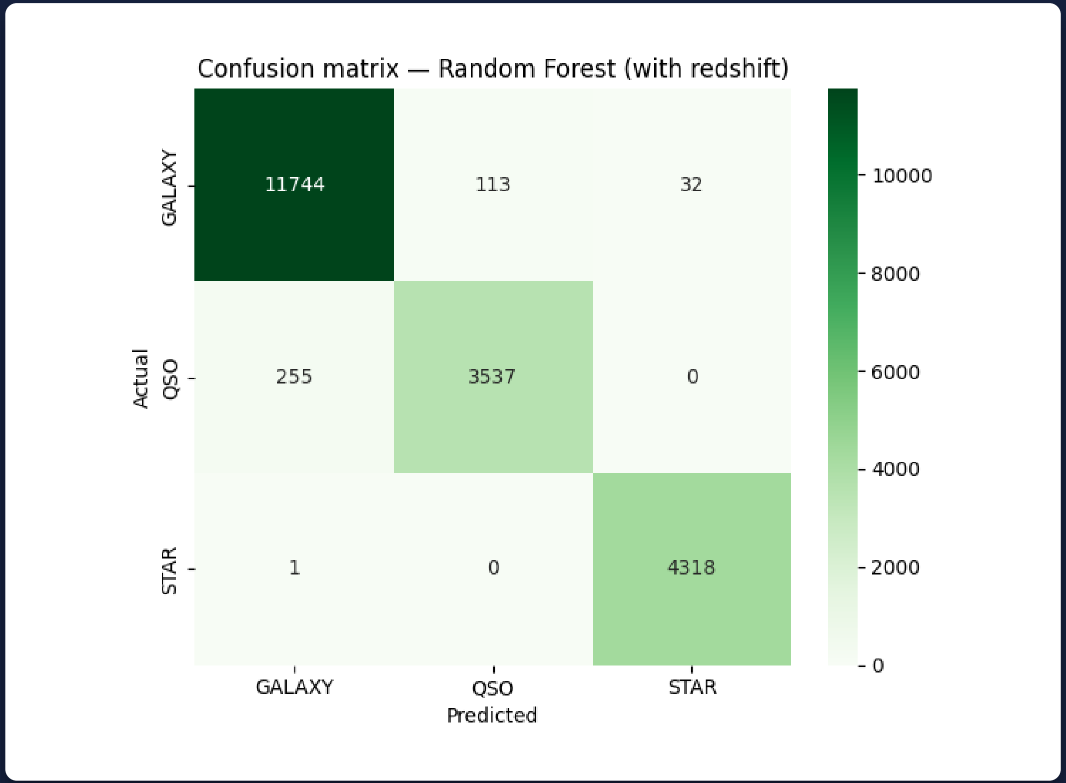
WHERE THE MODELS GET IT WRONG

Reading the mistakes

A confusion matrix shows not just how often each model is right, but which classes it confuses. The diagonal is correct; off-diagonal cells are errors.



Logistic Regression



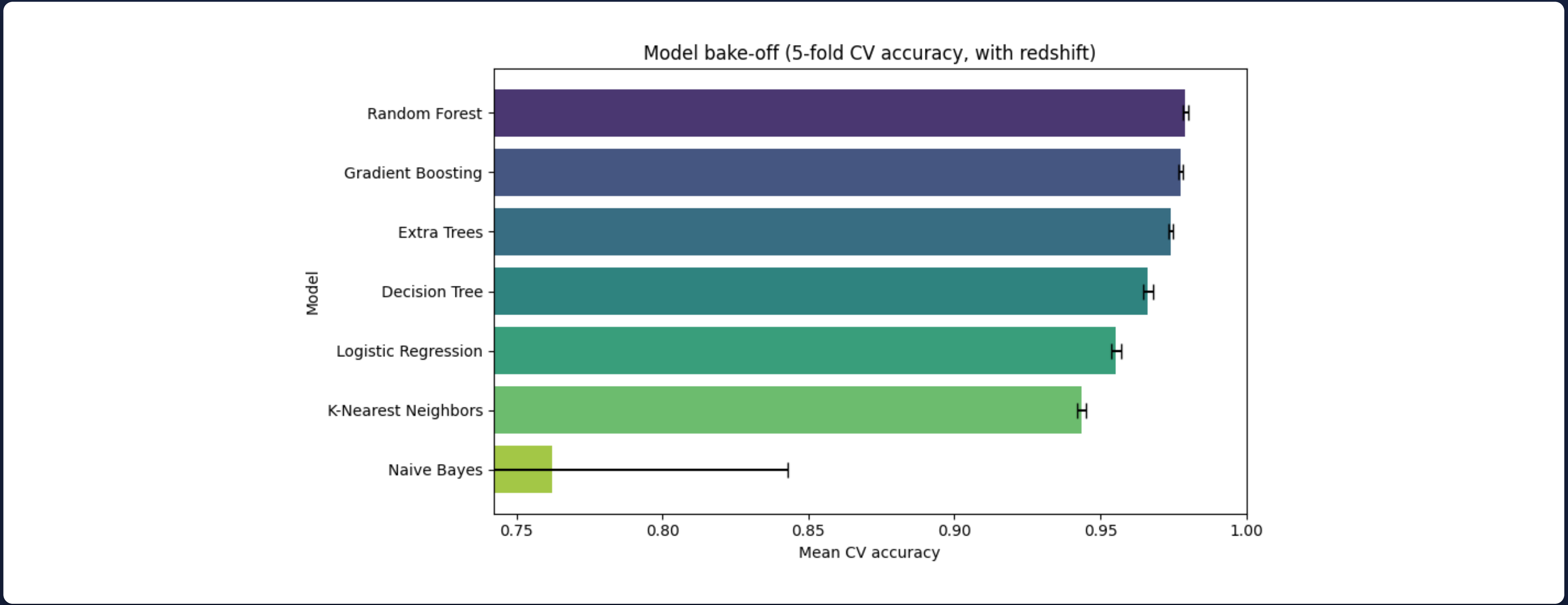
Random Forest

The pattern is physically sensible: *stars are almost never confused (redshift ≈ 0 makes them distinct), while nearly all remaining errors sit in the galaxy↔quasar cells — the genuinely hard cases. Random Forest tightens the diagonal further.*

CROSS-VALIDATED COMPARISON

Does a fancier model help?

Seven classifiers, 5-fold cross-validation, redshift included. The tree ensembles cluster at the top; Naive Bayes collapses because its independence assumption is violated by the correlated features.

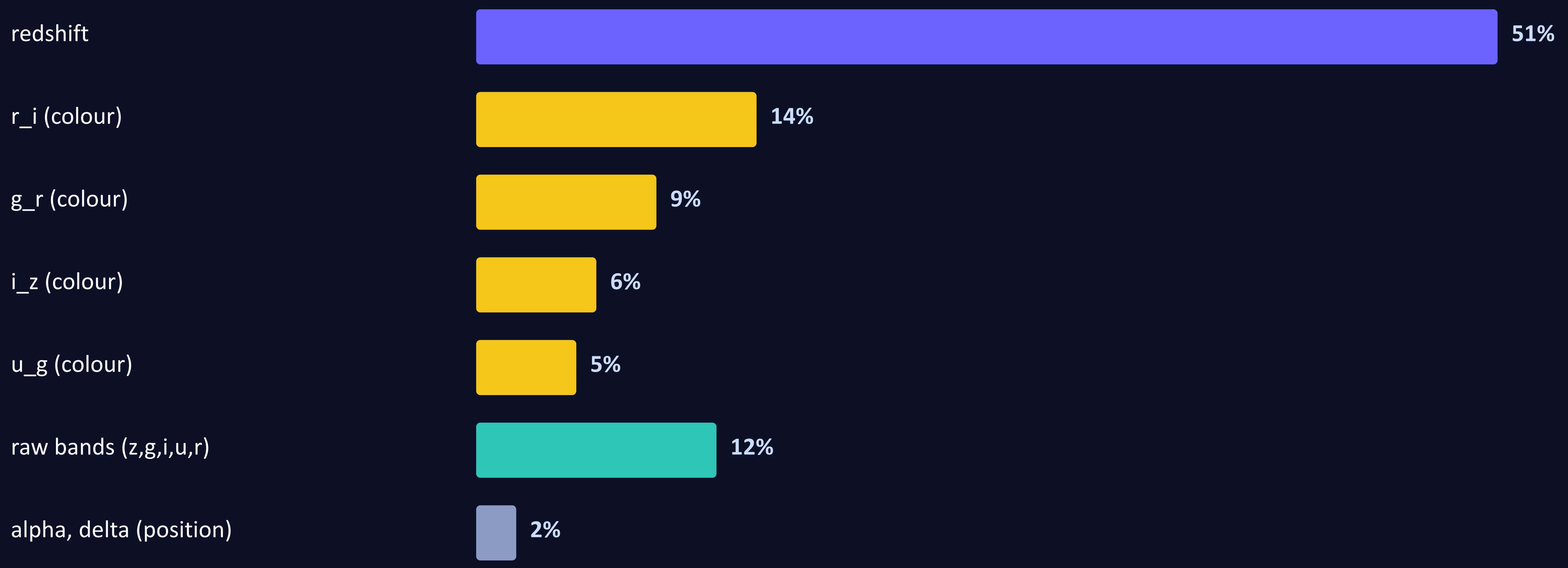


Once features are good, top ensembles converge — algorithm choice matters far less than beginners expect. Note Naive Bayes' wide error bar: unstable as well as inaccurate.

WHAT THE MODEL RELIED ON

The leakage, seen a second way

Random Forest reports how much it leaned on each feature. Redshift alone is over half the total — the same leakage story in a different picture.



Sanity check passed: *colours outrank the raw bands they came from, and sky position is ~0. The model independently recovered a century of astronomy.*

WHAT THIS PROJECT DEMONSTRATES

Takeaways that travel

Frame the real problem

The valuable task is classifying before spectroscopy — which changes which features are legitimate.

Interrogate every feature

Ask of each column: would I actually have this at prediction time? If not, its accuracy is a fiction.

Feature quality beats model choice

Engineered colours + honest inputs mattered more than which algorithm we picked.

Report honestly

88% stated plainly is worth more than a leaky 97% dressed up as success.

From raw light to a labelled sky

Exploratory → Descriptive → Feature Engineering → Predictive → Evaluation

The algorithm is the small part. The discipline — honest features, honest reporting — is the whole game.